

HandsOn-4

Please follow the instructions to complete the hands-on. Each instruction expect a command for the Git Bash.

1. Verify if master is in clean state.
2. Create a branch "GitWork". Add a file "hello.xml".
3. Update the content of "hello.xml" and observe the status
4. Commit the changes to reflect in the branch
5. Switch to master.
6. Add a file "hello.xml" to the master and add some different content than previous.
7. Commit the changes to the master
8. Observe the log by executing "git log --oneline --graph --decorate --all"
9. Check the differences with Git diff tool
10. For better visualization, use P4Merge tool to list out all the differences between master and branch
11. Merge the bran to the master
12. Observe the git mark up.
13. Use 3-way merge tool to resolve the conflict
14. Commit the changes to the master, once done with conflict
15. Observe the git status and add backup file to the .gitignore file.
16. Commit the changes to the .gitignore
17. List out all the available branches
18. Delete the branch, which merge to master.
19. Observe the log by executing "git log --oneline --graph --decorate"

Commands to add in created Repo

1 Configure Git to use VS Code for diffs and merges

```
git config --global diff.tool vscode
```

- Sets VS Code as the default tool for viewing diffs.

```
git config --global difftool.vscode.cmd "code --wait --diff $LOCAL $REMOTE"
```

- Defines how Git should launch VS Code to show side-by-side diffs.

```
git config --global merge.tool vscode
```

- Sets VS Code as the default tool for resolving merge conflicts.

```
git config --global mergetool.vscode.cmd "code --wait $MERGED"
```

- Defines how Git should launch VS Code for merge conflict resolution.
-

2 Create a new branch and add files

```
cd ~/GitDemo
```

- Change directory to your Git project.

```
git branch GitWork
```

- Create a new branch named GitWork.

```
git branch -a
```

- List all branches, including remote.

```
git checkout GitWork
```

- Switch to the new branch GitWork.

```
cat > hello.xml <<'EOF'
```

```
<root>Hello from branch</root>
```

```
EOF
```

- Create (or overwrite) hello.xml with specified content.

```
git add hello.xml
```

- Stage the new file to be committed.

```
git commit -m "Add hello.xml in GitWork branch"
```

- Commit the staged changes with a descriptive message.

git status

- Show the status of the working directory and staging area.
-

3 Switch back to master and create a conflict

git checkout master

- Switch back to the master branch.

```
cat > hello.xml <<'EOF'
```

```
<root>Hello from master</root>
```

```
EOF
```

- Create (or overwrite) hello.xml with different content on master.

git add hello.xml

- Stage the changed file.git commit -m "Add hello.xml in master branch"
 - Commit the changes on master branch.
-

4 View differences between master & GitWork

git diff master..GitWork

- Show text differences between master and GitWork branches.

git difftool master..GitWork

- Launch VS Code to visually compare differences between branches.
-

5 Merge GitWork into master (will cause conflict)

git merge GitWork

- Attempt to merge GitWork into master. Git will report conflicts in hello.xml.
-

6 Resolve conflict using VS Code mergetool

git mergetool

- Open VS Code merge editor to manually resolve conflicts in hello.xml.

After editing and saving the file:

git add hello.xml

- Mark the conflict as resolved by staging the file.

git commit -m "Resolve merge conflict in hello.xml"

- Commit the merge resolution.

7 Finish up

git log --oneline --graph --decorate

- Show a compact, graphical commit history including merges and branches.

git branch -d GitWork

- Delete the GitWork branch locally after successful merge.

git push origin master

- Push the merged changes on master to the remote repository.

Output Screens:

```

MINGW64/c/Users/LAVANYA/GitDemo
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git config --global diff.tool vscode
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git config --global difftool.vscode.cmd "code --wait --diff $LOCAL $REMOTE"
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git config --global merge.tool vscode
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git config --global mergetool.vscode.cmd "code --wait $MERGED"
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git branch Gitwork
fatal: a branch named 'Gitwork' already exists
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ AC
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git branch -d Gitwork
error: the branch 'Gitwork' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D Gitwork'
hint: Disable this message with 'git config set advice.forceDeleteBranch false'
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git branch -d Gitwork
error: the branch 'Gitwork' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D Gitwork'
hint: Disable this message with 'git config set advice.forceDeleteBranch false'
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git branch -a
* master
  remotes/origin/HEAD -> origin/master
  remotes/origin/master
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (master)
$ git checkout Gitwork
Switched to branch 'Gitwork'
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (Gitwork)
$ cat > hello.xml <<'EOF'
> <root> Hello from branch</root>
> EOF
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (Gitwork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time git touches it
LAVANYABLABTOP-7H755W1L MINGW64 ~/GitDemo (Gitwork)
$ git commit -m "Add hello.xml in Gitwork branch"
[Gitwork sdc392a] Add hello.xml in Gitwork branch
1 file changed, 1 insertion(+), 1 deletion(-)

```

```

MINGW64/c/Users/LAVANYA/GitDemo
LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitWork)
$ git status
On branch GitWork
nothing to commit, working tree clean

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitWork)
$ git checkout master
git: 'checkout' is not a git command. See 'git --help'.

The most similar command is
  checkout

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ cat > hello.xml <<'EOF'
<root>Hello from master</root>
> EOF

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time git touches it

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git commit -m "Add hello.xml in master branch"
[master dc2e436] Add hello.xml in master branch
1 file changed, 1 insertion(+), 1 deletion(-)

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git diff master..GitWork
diff --git a/hello.xml b/hello.xml
index abb0751..a0762bf 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
- <root>Hello from master</root>
+ <root> Hello from branch</root>

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git difftool master--GitWork
fatal: ambiguous argument 'master--GitWork': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git difftool master..GitWork

Viewing (1/1): 'hello.xml'
Launch 'vscode' [Y/n]? Y

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml

```

```

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master|MERGING)
$ AC

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master|MERGING)
$ git mergetool
Merging:
hello.xml

Normal merge conflict for 'hello.xml':
{local}: created file
{remote}: created file
hello.xml seems unchanged.
Was the merge successful [y/n]? Y

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master|MERGING)
$ git add hello.xml

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master|MERGING)
$ git commit -m "Resolve merge conflict in hello.xml"
[master fb77402] Resolve merge conflict in hello.xml

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git log --online --graph --decorate
* fb77402 (HEAD -> master) Resolve merge conflict in hello.xml

* 5de392a (GitWork) Add hello.xml in GitWork branch
* 9aa9466 Update hello.xml on GitWork
* 02f6f6d Add hello.xml on GitWork
* dc2e436 Add hello.xml in master branch
* 4eeea7d Add hello.xml on master with different content

* 1f7d96a (origin/master, origin/HEAD) Adding file1.txt and file2.txt
* 8d774af Delete file2.txt
* 42114d8 Delete file1.txt
* 8686a5d Added file1.txt and file2.txt to GitNewBranch
* e5af4ff Add .gitignore to ignore *.log files and log/ directories
* b4d7583 Added welcome.txt file Contains welcome message for Git training

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was 5de392a).

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (18/18), 1.72 KiB | 585.00 KiB/s, done.
Total 18 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 1 local object.
To https://github.com/Lavanya-Yannam/GitDemo.git

```

Lavanya-Yannam / GitDemo

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1 Branch

0 Tags

Go to file

Add file

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About

Lavanya-Yannam

Resolve merge conflict in hello.xml

fb77402 · 2 minutes ago

12 Commits

.gitignore	Add .gitignore to ignore *.log files and log/ directories	17 hours ago
file1.txt	Adding file1.txt and file2.txt	15 hours ago
file2.txt	Adding file1.txt and file2.txt	15 hours ago
hello.xml	Resolve merge conflict in hello.xml	2 minutes ago
welcome.txt	Added welcome.txt file	19 hours ago

README

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Lavanya-Yannam

Lavanya-Yannam / GitDemo

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file1.txt

file2.txt

hello.xml

welcome.txt

GitDemo / hello.xml

Lavanya-Yannam

Resolve merge conflict in hello.xml

fb77402 · 2 minutes ago

History

Code

Blame

2 lines (2 loc) · 53 Bytes

Raw

1

Hello from master branch!

2

Hello from Gitwork branch!